

$$\textcircled{1} \quad A = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\boxed{\text{Ug 15}}$$

a) Jacobi

b) GS

$$x^{(1)}, x^{(2)}, x^{(3)}$$

$$g(B)$$

a) Jacobi :

$$x_1^{(k+1)} = \frac{1}{2} (1 - 3x_2^{(k)}) = \frac{1}{2} - \frac{3}{2} x_2^{(k)}$$

$$x_2^{(k+1)} = \frac{1}{2} (3 - (-1)x_1^{(k)}) = \frac{3}{2} + \frac{1}{2} x_1^{(k)}$$

$$x^{(0)}$$

$$x^{(1)}$$

$$x^{(2)}$$

$$x^{(3)}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{7}{4} \\ \frac{7}{4} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{17}{8} \\ \frac{5}{8} \end{pmatrix}$$

$$A = D - E - F$$

$$A = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -3 \\ 0 & 0 \end{pmatrix}$$

$$B = D^{-1} (E + F)$$

$$= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & -3 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\frac{3}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$$

$$0 = \det(B - \lambda I) = \det \begin{pmatrix} -\lambda & -\frac{3}{2} \\ \frac{1}{2} & -\lambda \end{pmatrix}$$

$$= \lambda^2 + \frac{3}{4}$$

$$\Rightarrow \lambda^2 = -\frac{3}{4}$$

$$\lambda_{1,2} = \pm \frac{\sqrt{3}}{2} i$$

$$|\lambda_{1,2}| = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \rho(B) = \underline{\frac{\sqrt{3}}{2}} < 1 \Rightarrow \text{homogenisiert}$$

$$b) \quad \underline{GS} : \quad A = \begin{pmatrix} \underline{2} & \underline{3} \\ \underline{-1} & \underline{2} \end{pmatrix}, \quad b = \begin{pmatrix} \underline{1} \\ \underline{3} \end{pmatrix}$$

$$x_1^{(k+1)} = \frac{1}{2} (1 - 3 x_2^{(k)}) = \frac{1}{2} - \frac{3}{2} \underline{x_2^{(k)}}$$

$$x_2^{(k+1)} = \frac{1}{2} (3 - (-1) x_1^{(k+1)}) = \frac{3}{2} + \frac{1}{2} \underline{x_1^{(k+1)}}$$

$$x^{(0)} \quad x^{(1)} \quad x^{(2)} \quad x^{(3)}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} \frac{1}{2} \\ \underline{\frac{7}{4}} \end{pmatrix} \quad \begin{pmatrix} -\frac{17}{8} \\ \frac{7}{16} \end{pmatrix} \quad \begin{pmatrix} -\frac{5}{32} \\ \frac{91}{64} \end{pmatrix}$$

$$B_{GS} = (D - E)^{-1} F, \quad A = \begin{pmatrix} \boxed{2} & 3 \\ -1 & \boxed{2} \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 \\ -1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 0 & -3 \\ 0 & 0 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & -3 \\ 0 & 0 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 0 & -6 \\ 0 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\frac{3}{2} \\ 0 & -\frac{3}{4} \end{pmatrix}$$

$$\lambda_1 = 0, \quad \lambda_2 = -\frac{3}{4}$$

$$\Rightarrow p(B_{GS}) = \frac{3}{4} < 1$$

$\Rightarrow GS$ konvergiert